

Linear Algebra – MAT 2610

Section 6.3 (Orthogonal Projections)

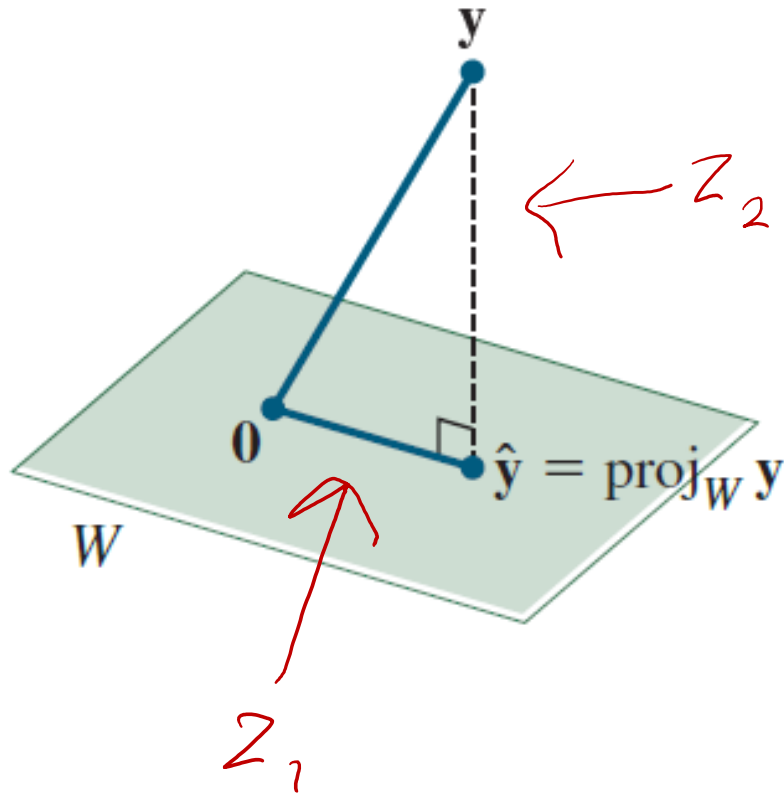
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Orthogonal Projection



Suppose W is a subspace of R^n .
For any vector y , we can write

$$y = z_1 + z_2$$

where z_1 is in W , and z_2 is
orthogonal to W .



Theorem 8 – Orthogonal Decomposition Thm

Let W be a subspace of R^n . Then each y in R^n can be written uniquely in the form

$$y = \hat{y} + z$$

where $\hat{y}$ is in W and z is in $W^\perp$. Moreover, if $\{u_1, \dots, u_p\}$ is an orthogonal basis of W , then

$$\hat{y} = \frac{y \circ u_1}{u_1 \circ u_1} u_1 + \dots + \frac{y \circ u_p}{u_p \circ u_p} u_p$$



Theorem 9 – Best Approximation Theorem

Let W be a subspace of R^n , $\hat{y}$ the orthogonal projection of y onto W .
Then $\hat{y}$ is the closest point in W to y , where

$$\|y - \hat{y}\| < \|y - v\|$$

For all v in W with $v \neq \hat{y}$



Theorem 10

If $\{u_1, \dots, u_p\}$ is an orthonormal basis for W , a subspace of R^n , $\hat{y}$ then

$$\hat{y} = \underbrace{(y \circ \underline{u_1})u_1 + \dots + (y \circ u_p)u_p}$$

If $\underbrace{U = [u_1 \ u_2 \ \dots \ u_p]}$, then

$$\hat{y} = \underbrace{UU^T y}$$



Example

If $u_1 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}$ and $u_2 = \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \\ 0 \end{bmatrix}$, find the orthogonal projection of $y = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$

Note that this is an orthonormal basis for a subspace of R^3 , so we can define $U = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & 0 \end{bmatrix}$, and $UU^T = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

This gives us:

$$\hat{y} = UU^T y = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}$$



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